

Earthquakes

Earth's surface is broken up into different sections, called tectonic plates. These are always on the move, and sometimes shift in ways that cause violent vibrations. Such vibrations are called earthquakes.

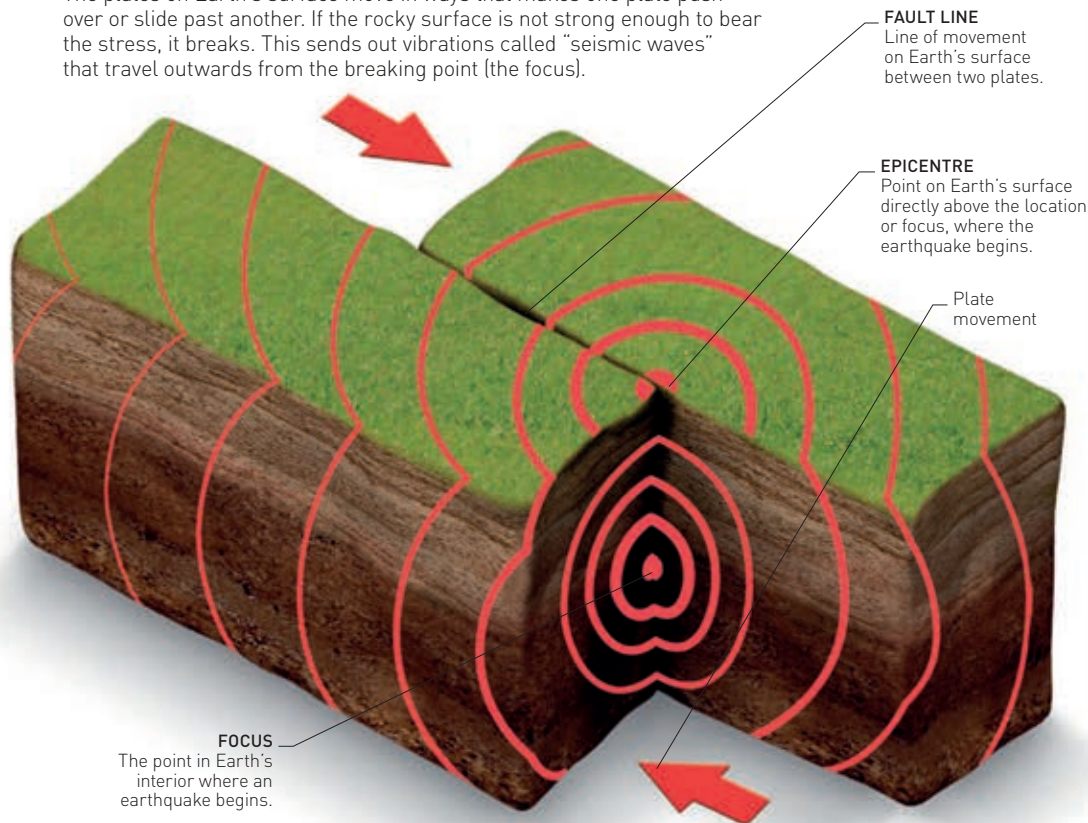
EARTHQUAKE-PRONE ZONES

Some countries are more affected by earthquakes than others because they lie on the boundaries of tectonic plates. The ten countries shown here have the highest death rates in the world due to violent earthquakes.



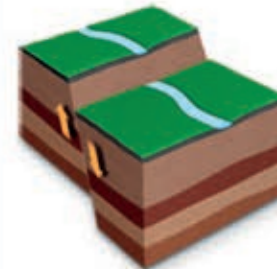
WHAT CAUSES EARTHQUAKES?

The plates on Earth's surface move in ways that makes one plate push over or slide past another. If the rocky surface is not strong enough to bear the stress, it breaks. This sends out vibrations called "seismic waves" that travel outwards from the breaking point (the focus).

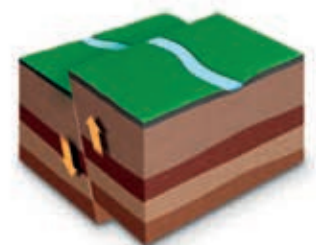


FAULT TYPES

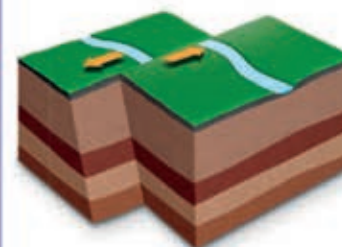
Faults are the boundaries between two moving tectonic plates. They are often the sites of earthquakes. The blocks of rock on either side of a fault can shift and slide past each other in various ways.



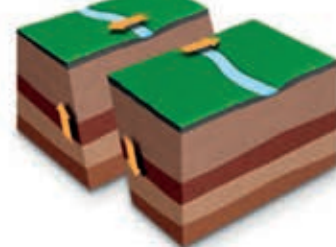
NORMAL FAULT
Rock on one side of the fault moves down, so it is lower than the rock on the other side of the fault.



REVERSE FAULT
One block is pushed up relative to the other, so it ends up at a higher level.



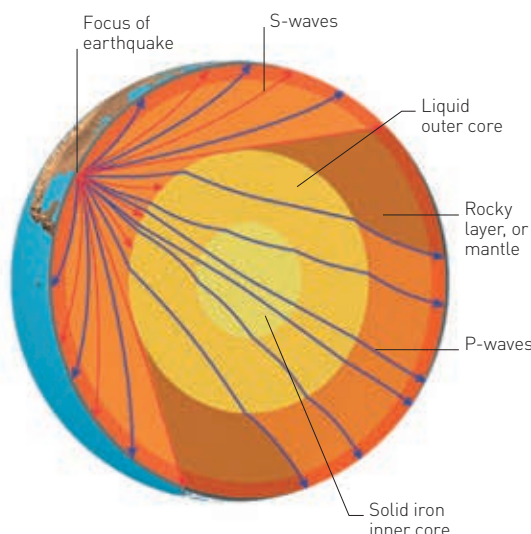
STRIKE-SLIP FAULT
The rocks on either side of the fault move in different directions, scraping side by side.



OBLIQUE-SLIP FAULT
The rocks on either side of the fault move sideways and up or down relative to each other.

SEISMIC WAVES

Two types of seismic waves created by an earthquake can travel right through Earth's interior. P-waves pass through both solid and liquid layers. S-waves, slower but more dangerous waves, move only through solid rock.



HOW SEVERE?

The Mercalli scale, below, is one way of measuring the intensity of an earthquake. For more precise estimates, scientists use the "moment magnitude" scale, which measures the amount of energy released during a quake.



I-II
Hardly felt by people, but can be measured by instruments.



III-IV
Felt indoors as a quick vibration that makes hanging objects shake.



V-VI
Rocking motion felt by people; also makes buildings tremble.



VII-VIII
Buildings shake badly, and tree branches break and fall.



IX-XI
Buildings crack and some fall; underground pipes torn apart.



XII
Most buildings are destroyed; rivers are forced to change course.

MAJOR EARTHQUAKES

Earthquakes can cause terrible devastation. The following have some of the highest-ever measurements on the moment magnitude scale.

- 1 CHILE, 22 MAY 1960**
Registering at magnitude 9.5, this is the largest recorded earthquake. It occurred in the Pacific Ocean and caused a series of tsunamis that left two million people homeless.
- 2 PRINCE WILLIAM SOUND, ALASKA, 28 MARCH 1964**
This huge earthquake (magnitude 9.2) caused a tsunami that rose to 67 m (220 ft) and hit Hawaii, Canada, and the USA. In the first day there were 11 aftershocks with magnitudes greater than 6.0.
- 3 NORTHERN SUMATRA, 26 DECEMBER 2004**
This ruptured the longest fault of any recorded quake, spanning 1,500 km (900 miles) in ten minutes. More than 227,000 people were killed by the resulting tsunami.
- 4 HONSHU, JAPAN, 11 MARCH 2011**
This 9.0-magnitude earthquake occurred off the coast of Japan and reached depths of 24.4 km (15 miles). The resulting tsunami caused more than 15,800 deaths.
- 5 KAMCHATKA, RUSSIA, 4 NOVEMBER 1952**
Registering a magnitude of 8.2, this earthquake set off a Pacific-wide tsunami that hit Peru, Chile, New Zealand, many Pacific islands, and California, USA.